

# AY25 PROGRAM ASSESSMENT REPORT

v2026.1

## Tracking Data

|                                                                              |                                        |
|------------------------------------------------------------------------------|----------------------------------------|
| CIM code (e.g., ME-CM)                                                       | MS-CE                                  |
| Program Name                                                                 | Master of Science in Civil Engineering |
| Assessment Year                                                              | 2024-2025                              |
| Program Assessment Coordinator                                               | Mehdi Modares, Brent Stephens          |
| Year of Current Assessment Plan                                              | 2024-2025                              |
| No Assessment This Year <b>per Plan</b>                                      | <input checked="" type="checkbox"/>    |
| LO(s) Assessed                                                               | 0                                      |
| Presented at program authority meeting date:<br><i>Please attach minutes</i> | n/a                                    |

## Methodology

The Master of Science in Civil Engineering is awarded upon demonstration of an ability to comprehend and contribute to practice and knowledge in construction, structural, or transportation engineering. The program generally consists of 9-12 credit hours of required core courses in the student's field of study, 12-15 credit hours of elective courses (a mix of technical and general electives), and 6 credit hours for research and thesis.

Research is conducted under the supervision of a full-time tenured or tenure track faculty member, and students should work to involve all members of their research committee. To successfully fulfill the requirements for the degree, students must draft a thesis document and pass an oral examination in defense of the thesis, which is given as an open university seminar. The thesis defense must meet with the approval of the research committee; if it does not, the committee has the authority to determine whether or not to grant a re-examination.

**Learning objectives for this program are for students to:**

1. Demonstrate knowledge exceeding the fundamentals gained in undergraduate work.
2. Demonstrate the skills necessary to engage in and conduct research.
3. Demonstrate the ability to effectively communicate technical information (oral and written).

To assess these learning goals, the program uses student performance in the thesis defense exam and assessment of the quality of thesis. Both are assessed on the basis of knowledge demonstration and effective communication. The CAEE Department has initiated a set of rubrics for assessing the thesis defense exam, which are shown in the appendix of this document.

## Results

### Data table

In Fall 2024, the program had 8 students enrolled. Of those 8 students, 5 later converted to a coursework-focused master's degree (i.e., M.E. or M.A.S.) rather than the research-focused M.S. degree, at some point during the academic year. Among the 3 remaining students in the program, 2 are focused in the structural engineering concentration and 1 is focused in the construction engineering and management concentration.

### Description of findings

No students completed the MS-CE degree and thesis defense in the 2024-2025 academic year, so no assessment is performed this year.

### Prior CI plan impact

In the previous assessment reports, institutional reviewers made several recommendations to improve upon our assessment plans, including converting to an updated format, indicating more clearly which artifacts are used and collected, and to improve upon our CI plans.

### This cycle's CI plan

The department has initiated a data collection process using rubrics to assess performance via thesis defenses and will analyze data retrospectively in cohorts of multiple years to increase sample sizes.

Additionally, given the large proportion of students who entered the M.S. program but later transferred to a coursework-focused professional master's program, we have had ongoing discussions with our graduate admissions team regarding the imperative of accepting only a select few students who clearly demonstrate a propensity for and interest in research into the M.S. program, steering all others to coursework-based professional master's programs to limit this kind of confusion among students and advisors.

### IIT CAEE – Master of Science Thesis Defense Examination Evaluation Rubric

**Instructions:** The student submits a written thesis and then presents the thesis orally. The committee, comprising at least two full-time faculty members in the department, evaluates the work on its technical foundation, methods and hypotheses, novelty, and clarity. The defense exam can also include general questions about any material relevant to the research. The defense exam committee will then complete the following evaluation rubric for each candidate taking the thesis defense examination using a consensus approach.

Student: \_\_\_\_\_ Program (CE, ARCE, ENVE): \_\_\_\_\_ Exam Date: \_\_\_\_\_

| Criterion                                                   | Below Expectations (1)                                                                                                                                                                                                    | Meets Expectations (2)                                                                                                                                                                                                                                 | Exceeds Expectations (3)                                                                                                                                                                                                                                           | Score (1-3) & Comments |
|-------------------------------------------------------------|---------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|------------------------|
| 1. Motivation and background                                | The reasons for the research are not covered or only minimally covered. Minimal background research to date. Lacking context in existing literature. Limited or no signs of critical thinking regarding prior literature. | Big picture presented. Reasons for research questions laid out. Relevant prior work is cited and understood. Some command of existing literature is shown.                                                                                             | Motivation is clear and documentation and/or data is used to show the importance and need for the research. Relevant papers and work are cited, understood, and evaluated critically. Strong command of existing literature is demonstrated.                       |                        |
| 2. Methods development, analysis, and discussion of results | Methods are not well developed, understood, or communicated. Testable hypotheses are not well developed. Analysis is lacking and/or does not display mastery of fundamental concepts.                                     | Methods are understood and evaluated with coherent analysis and discussion. Limitations are identified, with some assistance needed. Testable hypotheses are developed and utilized. Research demonstrates good understanding of fundamental concepts. | Methods are well understood and critically evaluated. Limitations are well defined. Testable hypotheses are well developed, analyzed, and lead to novel findings. Research builds on a complete understanding of fundamental concepts and mastery is demonstrated. |                        |
| 3. Written communication                                    | Limited command of language and vocabulary. Grammatical mistakes. Poor thesis organization. Figures are poor quality.                                                                                                     | Demonstrates good command of language and vocabulary. Minimal grammatical mistakes. Thesis is organized. Figures are readable.                                                                                                                         | Full command of language and vocabulary. No grammatical mistakes. Thesis is well organized and complete. Figures are publication quality.                                                                                                                          |                        |
| 4. Oral communication and presentation                      | Limited command of language and vocabulary. Speaker cannot be heard clearly. Slides and thesis are difficult to read. Organization poor. Grammatical or spelling mistakes.                                                | Command of language and vocabulary, with some clarification needed. Speaker projects well. Slides and thesis are clear. Presentation is organized.                                                                                                     | Speaker projects well and demonstrates excellent command of language and vocabulary. Slides and thesis are superb. Presentation is superb. Speaker is on par with speakers at national conferences.                                                                |                        |
| 5. Responses to questions                                   | Does not understand questions. Is not able to answer questions.                                                                                                                                                           | Understands questions. Answers questions, potentially with clarifications.                                                                                                                                                                             | Understands and responds to questions as well as gives context to larger issues around questions.                                                                                                                                                                  |                        |

Exam Result (Pass/Fail): \_\_\_\_\_

Overall Score (out of 15): \_\_\_\_\_

**Committee Members:**

**Additional Comments:**